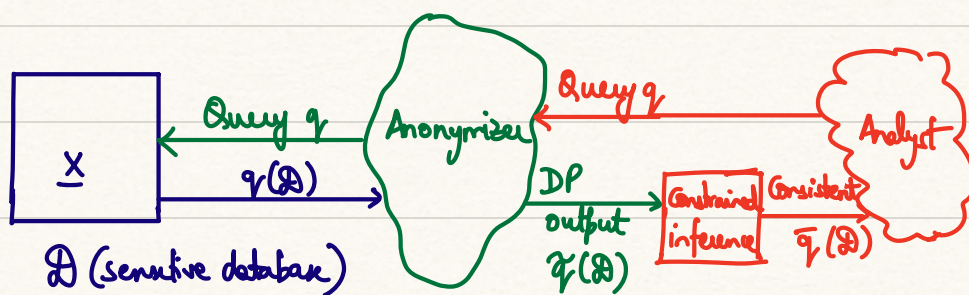


Lecture: DP in practice -III: Consistent histograms

In this lecture, we discuss an important component of the privacy pipeline that allows the outputs of DP algorithms to be used by downstream applications: post-processing for consistency. "Consistency" is a generic term used to refer to the requirement that the DP estimator of a statistic respect the theoretical properties of the statistics such as non-negativity, monotonicity, etc.

Such consistency constraints were famously put in place in the TopDown algorithm used for releasing DP statistics (histograms) in the 2020 US Census. (see [Abowd et al., "Census TopDown: DP data, incremental schemas, and consistency with public knowledge," 2019]). We now discuss an algorithm for post-processing histograms for consistency, with a view towards answering "range queries" accurately [Hay, Rastogi, Miklau, Suciu, "Boosting the accuracy of DP histograms through consistency," 2010].

Pictorially, the data privatization "pipeline" has the following structure:



Suppose that the query q is the histogram $f_{\text{hist}}(\underline{x})$. We work with the "add/remove" setting of DP (as indicated earlier, one can transition from the add/remove model to the "replace" model, with a factor-of-2 loss in privacy budget).

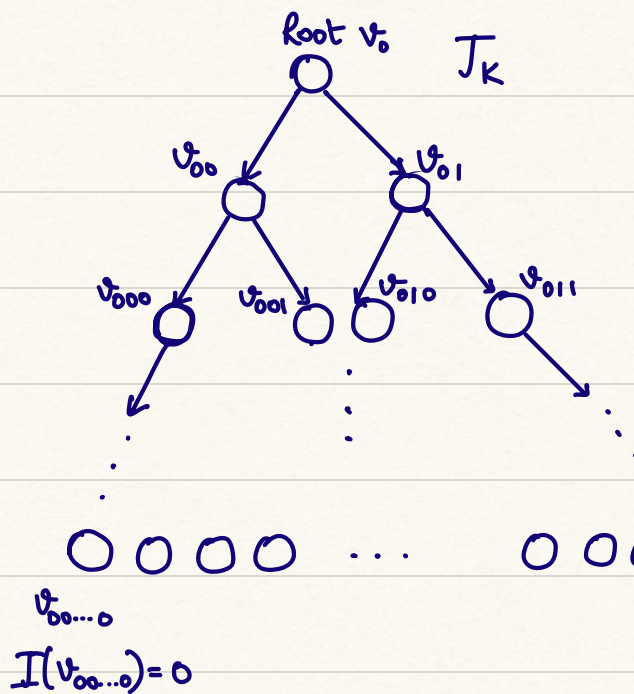
We assume that the dataset $\underline{x} \in \{0, 1, \dots, K-1\}^N$, for some integer $K \geq 1$. Recall that we have

$$\Delta_{f_{\text{hist}}} = 1. \quad [\text{why?}]$$

Recall also the vanilla ϵ -DP Laplace mechanism for this statistic:

$$M_{\epsilon}^{\text{Lap}}(\underline{x}) = f_{\text{hist}}(\underline{x}) + \left(\text{Lap}\left(\frac{1}{\epsilon}\right)\right)^{\otimes K}.$$

However, our interest will be in the binary tree mechanism (recall!) for releasing $f_{\text{hist}}(\underline{x})$: for simplicity, assume that $K = 2^l$, for some $l \geq 1$. One then induces a binary tree on the leaves that are the bins $\{0, 1, \dots, K-1\}$, with the intervals $I(v)$ associated to a leaf node v being the corresponding bin and the interval associated with any non-leaf node u being the union of the intervals of its children.



To any node $u \in J_K$, add $\text{Lap}(\frac{1}{\epsilon})$ noise to the count in its assoc. interval:

$$Y(u) = X(u) + Z_u$$

Construct

$$(M_{\epsilon}^{\text{ST}}(x))_i = \sum_{u \in C(i)} Y(u).$$

↓ covering of $i \in \{0, 1, \dots, K-1\}$

A histogram estimate is obtained by taking differences of adjacent coordinates of the CDF. Our interest is in **post-processing** strategies that optimize the ℓ_2 -error while ensuring consistency. The consistency constraints P that we care about are simply:

(i) Any estimate $\hat{f}_{\text{hist}}$ of f_{hist} must be non-negative.

(ii) We must have that the count estimate for any node $u \in J_K$ is equal to the sum of count estimates of its children.

Let $\hat{f}_{\text{hist}}$ be a post-processed estimate of $M_{\epsilon}^{\text{ST}}(x)$ that respects the consistency properties P above with associated node estimates $(W(u) : u \in J_K)$.

We are interested in minimizing $\|W - Y\|_2^2$.

where $X = (X_u : u \in J_K)$.

For any $v \in T$, let $z(v)$ be defined recursively as:

$$z(v) = \begin{cases} y_v, & \text{if } v \text{ is a leaf node,} \\ \frac{2^l - 2^{l-1}}{2^l - 1} \cdot y_v + \frac{2^{l-1} - 1}{2^l - 1} \cdot \sum_{u \in \text{child}(v)} z(u), & \text{o.w.} \end{cases}$$

[Hay et al. (2010)] argue that the following (recursively defined) estimator W minimize $\|W - X\|_2$:

$$W(v) = \begin{cases} z(v), & \text{if } v \text{ is the root,} \\ z(v) + \frac{1}{2} \left(W(P(v)) - \sum_{u \in \text{child}(v)} z(u) \right), & \text{o.w.} \end{cases}$$

Proof of Claim: We first show that $W(r)$ corresponds to the optimum consistent estimate at the root r . But before that, we introduce some notation: let $z(\text{succ}(v)) \triangleq \sum_{u \in \text{child}(v)} z(u)$.

Recall that our optimization problem of interest is:

$$\text{minimize } \sum_{v \in T_K} (W(v) - y(v))^2$$

$$\text{s.t. } \sum_{u \in \text{child}(v)} W(u) = W(v), \quad \forall v \in T_K.$$

In other words, we wish to solve the following unconstrained optimization problem:

$$\text{minimize } \sum_{v \in T_k} \left(\sum_{u \in L(v)} W(u) - Y(v) \right)^2,$$

where $L(v) \triangleq$ leaves in the subtree with v as the root. Taking the derivative w.r.t. $W(u)$ and setting it to 0, we obtain after some algebra that for all leaves $u \in T_k$, we must have

$$\sum_{v: u \in L(v)} W(v) = \sum_{v: u \in L(v)} Y(v). \quad \text{---(1)}$$

Summing over all leaves $u \in T_k$ and interchanging the order of summations, we obtain from (1) that

$$\sum_v \sum_{u \in L(v)} W(v) = \sum_v \sum_{u \in L(v)} Y(v). \quad \text{---(2)}$$

Consider now the LHS of (2). Observe that [with $H(r) = l-1$ and $H(u) = 0$, for a leaf u]

$$\begin{aligned} \sum_v \sum_{u \in L(v)} W(v) &= \sum_{i=0}^{l-1} \sum_{v: H(v)=i} \sum_{u \in L(v)} W(v) \\ &= \sum_{i=0}^{l-1} \sum_{v: H(v)=i} 2^i \cdot W(v) \\ &= \sum_{i=0}^{l-1} 2^i \cdot W(n) = (2^l - 1) \cdot W(n). \quad \text{---(3)} \end{aligned}$$

Next, consider the RHS of (2). We have that [by III^a reasoning as above]

$$\sum_v \sum_{u \in L(v)} Y(v) = \sum_{i=0}^{l-1} 2^i \cdot \sum_{v: H(v)=i} Y(v). \quad \text{---(4)}$$

Reconciling (2), (3), and (4), we get that

$$W(r) = \frac{1}{2^l - 1} \cdot \sum_{i=0}^{l-1} 2^i \cdot \sum_{v: H(v)=i} Y(v),$$

and it can be checked hence that $W(r) = Z(r)$.

For nodes v other than the root, assume that we have computed $W(u)$, for $u = P(v)$. It can be argued that $W(v)$, for fixed $W(u)$, is independent of $W(a)$, for any node 'a' not in the subtree with u as the root.

For nodes $a \in \text{Subtree}(u)$, the optimization problem reduces to:

$$\text{minimize } \sum_{a \in \text{Subtree}(u)} (W(a) - Y(a))^2$$

$$\text{s.t. } \sum_{b \in \text{child}(a)} W(b) = W(a), \quad \forall a \in \text{Subtree}(u),$$

$$\text{and } \sum_{v \in \text{child}(u)} W(v) = W(u).$$

Again, the above optimization problem can be reduced to:

$$\text{minimize } \sum_{a \in \text{Subtree}(u)} \left(\left(\sum_{v \in L(a)} w(v) \right) - y(a) \right)^2$$

$$\text{s.t. } \sum_{v \in L(u)} w(v) = w(u).$$

Employing Lagrange multipliers and the KKT conditions, it can be shown that

$$w(v) = z(v) + \frac{1}{2} \left(w(u = \mathcal{P}(v)) - z(\text{succ}(v)) \right). \quad \square$$